A Study on Computational Metrics

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Abstract

This report demonstrates essential LaTeX features including structured sections, equations, tables, figures, algorithms, custom macros, and bibliography management. We illustrate their usage through a new context with different formulas.

1 Introduction

Mathematical relationships are fundamental in computational studies. For example, the Gaussian integral is given by

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}, \tag{1}$$

which we will reference in later discussions.

2 Methods

We measured the refractive index of the material using the custom macro

$$n = \frac{c}{v}, \tag{2}$$

where c is the speed of light in vacuum and v is the speed of light in the material.



Figure 1: Sample illustration figure.

2.1 Figures

Figure 1 shows a demonstration image included to illustrate figure handling.

2.2 Tables

Table 1 shows sample measurements of sensor readings.

Sensor	Reading 1	Reading 2	Reading 3
A	12.5	13.1	12.9
B	14.0	13.8	14.2
C	11.8	12.0	11.9

Table 1: Sample sensor readings.

3 Algorithm

Algorithm 1 demonstrates computing the factorial of a number.

$n \leftarrow$ input integer
$result \leftarrow 1$
for $i \leftarrow 1$ **to** n **do**
    $result \leftarrow result \times i$
end for

Algorithm 1: Compute factorial of an integer.